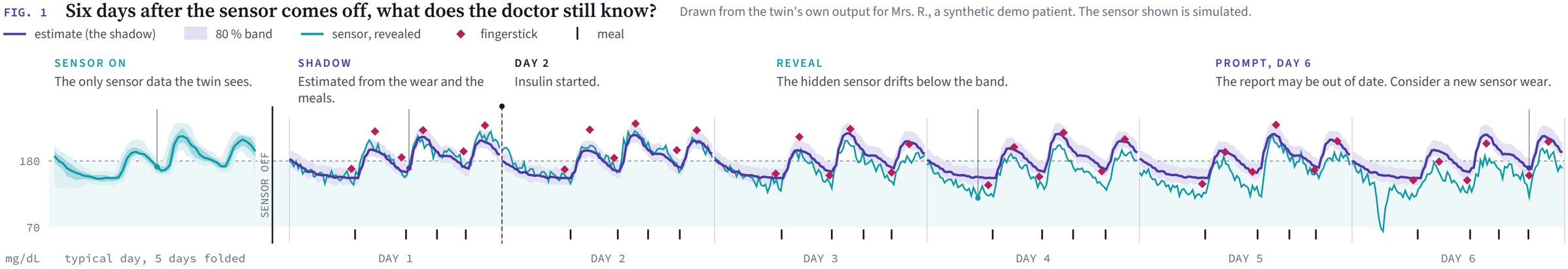
# Chhaya: the diabetes digital twin that keeps going when the sensor is off.

It fuses a patient's health record with a wearable glucose sensor, draws an honest band around every estimate, says when its report may be out of date, and proves every claim on patients it never saw.

THIS PAGE ANSWERS THE BRIEF

1. Static stream
2. Dynamic stream
3. Fusion of the two
4. Adverse-event prediction
5. Doctor dashboard

**FIG. 1 Six days after the sensor comes off, what does the doctor still know?** Drawn from the twin's own output for Mrs. R., a synthetic demo patient. The sensor shown is simulated.




## 01 SOURCES

CC BY 4.0

**ShanghaiT2DM**
100 patients

CC BY-NC-SA 4.0

**CGMacros**
45 participants

SYNTHETIC

**Mrs. R.**
demo patient

**OPEN OR SYNTHETIC ONLY**
No credentialed data. Never redistributed.

## 02 TWO STREAMS

STATIC ① **Health record**
FHIR-shaped

Diagnosis, duration
HbA1c, fasting glucose
BMI, age, sex
Medications
Genetic marker: synthetic, read by no model

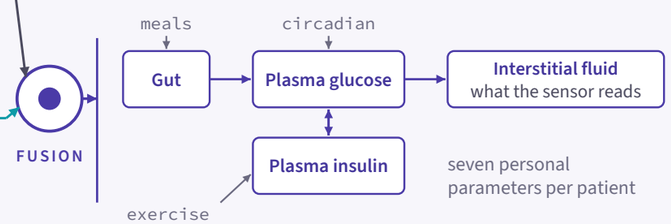
DYNAMIC ② **Time series**

- **CGM sensor**, wear only
- **Meal log**, time and carbs
- **Wrist band**, METs
- **Fingersticks**, after the wear

## 03 THE TWIN

### Personalised glucose–insulin model ③

Prior from the record. Evidence from the sensor wear. Output: a shadow with an 80 % band.




E-DES physiology (Maas et al., 2015), JAX, float64.

HANDS ON: shadow + 80 % band · fingerstick filter · drift prompt

④ EVENT MODEL, RESEARCH

Logistic regression predicts a post-meal excursion (two readings above 180 mg/dL within 2 h). **AUROC 0.82** with the sensor, **0.71** without.

## 04 THE GATE

Nothing reaches the screen unless it was scored **once** on unseen patients.

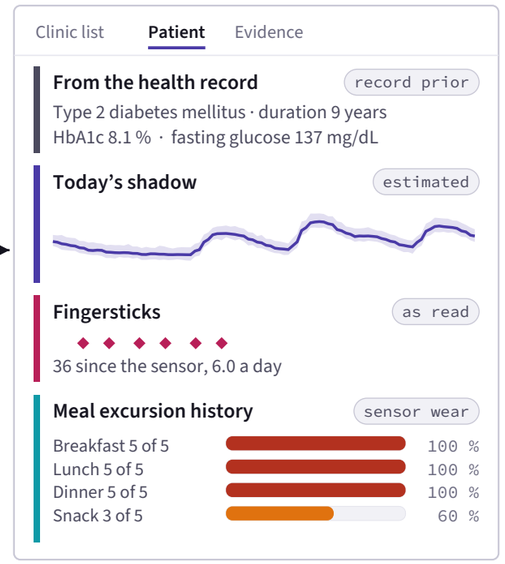
**5 passed · 4 missed**

Bars written first. All nine published.

| | | |
|---|---|---|
| Shadow: closer than the average day | PASSED | 1.4 mg/dL, small |
| Fingersticks keep the report truer | PASSED | at about six a day |
| Prompt flags a stale report | DESCRIPTIVE | AUROC 0.82 |
| Record improves the estimate | MISSED | no detectable gain |
| Fused alert is usable | MISSED | slope 0.48; 8.9 false alerts per patient-week |

## 05 THE SCREEN

### ⑤ The doctor's screen




One panel per stream, plus the shadow.
Never shown: dose advice, low-glucose estimates, alarms.

## Every box on this page is a folder in the repository. Every number is a command.

`src/chhaya/` data · twin · eval · product · dashboard

Static stream · Dynamic stream · Twin · Evaluation gate · Research only · Data flow

Research prototype. Not a medical device. Not for clinical decisions.

# Technical stack and AI/ML model card

Every component, library and method behind the shadow, the band, the prompt and the event model. Details were checked against the code and the decision records.

## A TECHNICAL STACK

| LAYER | TECHNOLOGY | ROLE |
|---|---|---|
| **Language and environment** | Python 3.12 to 3.13; uv with a locked environment | Reproducible install from one lockfile |
| **Numerical core** | JAX 0.11 (float64), NumPy 2.5, SciPy 1.18 | ODE integration, ensembles, optimisation |
| **Machine learning** | scikit-learn 1.9, LightGBM 4.7 | Logistic-regression arms; boosted-tree check |
| **Data handling** | pandas 3.0, PyArrow 25, openpyxl, xlrd, requests, tabulate | Loaders; checksum-verified download |
| **Interoperability** | FHIR-shaped Bundle: Patient, Condition, Observation with LOINC codes, MedicationStatement | The static record stream |
| **Dashboard** | Python standard-library HTTP server; HTML, CSS and vanilla JavaScript; pre-computed JSON bundle | Offline clinician interface; static export |
| **Quality** | pytest (400+ tests), ruff | Leakage, wording and parity tests |
| **Typography** | Source Sans 3, Source Serif 4, Source Code (SIL OFL 1.1) | Interface fonts, licence included |

## B AI/ML MODEL CARD

| COMPONENT | METHOD AND SETTINGS |
|---|---|
| **Method family** | Physics-based Bayesian digital twin plus classical machine learning. **No neural network, no pretrained model, no large language model.** |
| **Physiology** | E-DES glucose–insulin model (Maas et al., 2015) in the CGM-calibrated form of Erdős et al.: gut, plasma glucose, plasma insulin and interstitial fluid, with a circadian term on hepatic output and an exercise term on insulin-dependent uptake. Seven personal parameters. RK4 on a 1-minute grid in JAX, float64. |
| **Fusion and calibration** | The health record sets a prior on basal glucose (its fasting glucose). The sensor wear is the likelihood. MAP fit by SciPy least squares from six starts. |
| **Uncertainty** | Laplace posterior, widened for misfit and autocorrelated residuals. A 200-member ensemble (member 0 is the MAP) gives the 80 % band. |
| **Estimate** | An equal-weight blend (0.5) of the physiology and the patient's own average day. |
| **Fingersticks** | A linear-Gaussian filter on the deviation from the daily shape. The deviation fades with a 120-minute time constant, chosen on development patients. Live use reads only earlier fingersticks; the retrospective report uses an exact smoother. Meter readings are converted to the sensor's scale first. |
| **Drift prompt** | CUSUM on standardised fingerstick surprises, allowance 0.5 SD per fingerstick. The threshold is set on development recordings so that at most 10 % of stable recordings raise it. |
| **Event model** | L2 logistic regression (C = 1.0) with median imputation and standardisation on five input sets: 13 record inputs, sensor-week history, fingerstick features, all three fused, and a live sensor reading. Patient-grouped cross-validation; LightGBM at library defaults as a check. Label: two consecutive readings above 180 mg/dL within 120 minutes of a meal start. |
| **Validation** | Patient-level split: one half tunes, the other half is scored once. Bars written first. 95 % intervals from a 2,000-resample bootstrap over patients. |
| **Intended use and limits** | A research prototype for a clinician reviewing a past sensor report. The physiology alone beats the average day in only 60 % of held-out patients; the blend is what wins. Evidence comes from free-living adults of mixed glycaemic status and from supervised-care patients, with no Indian patient data. Not for dosing. |

## C DATA SOURCES AND LICENCES

| DATASET | LICENCE | CONTENT AND ROLE |
|---|---|---|
| **ShanghaiT2DM** | CC BY 4.0 | 100 patients with type 2 diabetes, 109 recordings, supervised care. CGM, fingersticks about six a day, labs and medications. |
| **CGMacros** | CC BY-NC-SA 4.0 | 45 free-living participants. CGM, wrist band, meals and bloodwork. |
| **Mrs. R.** | Synthetic | One demo patient built in code and tagged synthetic inside her own record. The committed demo holds only her. |

No credentialed data. Datasets are fetched by a checksum-verified script and never redistributed.

## D HOW ONE ESTIMATE IS MADE

**1 Health record**
Fasting glucose sets the basal-glucose prior

→

**2 Sensor wear**
Calibration window of k days; MAP fit from six starts

→

**3 Laplace posterior**
200 parameter sets; member 0 is the MAP

→

**4 Simulate**
E-DES, RK4, 1-minute grid. Inputs: meal log, wrist band

→

**5 Blend**
0.5 physiology + 0.5 the patient's own average day

→

**6 Fingerstick filter**
120-minute fade; converted to the sensor's scale

→

**7 Output**
Shadow with an 80 % band; CUSUM drift prompt

**EVENT MODEL, RESEARCH** 13 record inputs, sensor-week history and fingerstick features go into a standardised L2 logistic regression that scores the chance of a post-meal excursion. It is stopped at the gate as an alert: calibration slope 0.48 and 8.9 false alerts per patient-week.

# Evidence, contracts and governance

Every box has a contract. Every claim has a record.

THE SCOREBOARD

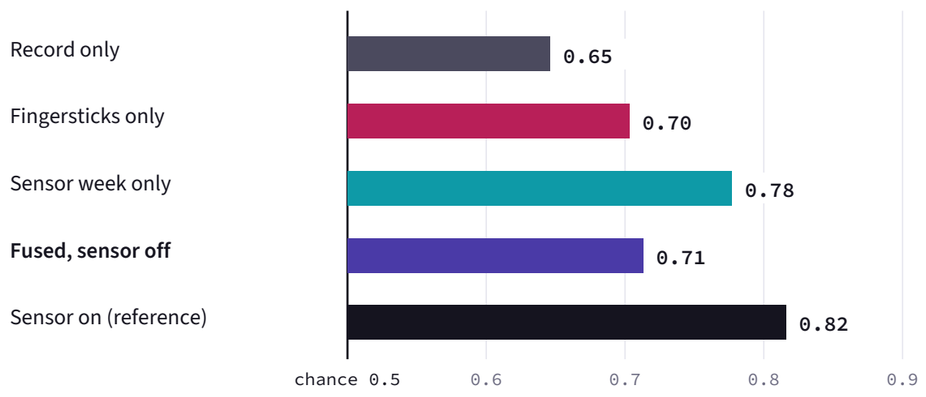
**5 passed · 4 missed · all published**

## 01 MODULE CONTRACTS

| MODULE | TAKES IN | HANDS ON |
|---|---|---|
| `data/` | Two open datasets and the synthetic patient. Checksum-verified download, never redistributed. | Recordings: sensor trace, meal log, wrist band, fingersticks and record fields. A label check of sensor lows and highs against fingersticks. |
| `twin/` | The wear window, the record's prior and the meal log. | A calibrated personal model: the shadow with an 80 % band, the fingerstick-nudged estimate, and the drift prompt. |
| `eval/` | Recordings and the bars written beforehand. | Scored-once results: `summary.json`, `report.md` and `provenance.json` (commit and settings of the run). A second run is refused. |
| `product/` | Results and decision records. | A claims registry; Evidence-screen wording tested word for word against the records; the FHIR-shaped record and its summary, the meal excursion history, the synthetic patient and the what-if. |
| `dashboard/` | A pre-computed JSON bundle. | Three screens that run offline. The patient screen has one panel per stream, plus the shadow. A static export for hosting. Nothing is fitted while it runs. |

## 02 MORE STREAMS WERE NOT BETTER

Post-meal excursion predicted at meal time, 47 held-out patients. AUROC by what the model is given.




The fused model sits **below the sensor week alone**. AUPRC: fused 0.596, sensor week 0.626, fingersticks 0.642, against an event rate of 0.39. Calibration slope for the fused model: 0.48, so it is not used as an alert.

## 03 THE PROOF PROTOCOL

1. **Write the bars first**
   Each bar and its comparator is dated in `docs/PREREGISTRATION.md` before the run. Amendments are dated too.
2. **Split by patient**
   One half of every cohort tunes every setting. The other half is not touched.

   
   
3. **Freeze the code**
   The run is made from committed code. `provenance.json` records the commit and the settings.
4. **Score once**
   Calibrate on k days, hide the rest, estimate, reveal. The command needs `--confirm` and refuses a second run.
5. **Count the uncertainty**
   95 % intervals from a bootstrap over patients (2,000 resamples in Gate 3).
6. **Publish everything**
   Misses included. Every claim on the Evidence screen is tested word for word against a decision record.

## 04 RESULTS AT TRUE SIZE

| RESULT | COHORT | NUMBER | OUTCOME |
|---|---|---|---|
| **The reveal** | 19 held-out CGMacros participants, free-living, 7 with type 2 | 1.4 mg/dL (6 %) closer than the patient's average day | Passed, small |
| **Fingersticks keep the report truer** | 29 held-out ShanghaiT2DM patients, supervised care, about six a day | 2.8 mg/dL closer as they arrive; 0.3 at one a day | Passed |
| **Excursions, sensor off** | 47 held-out ShanghaiT2DM patients | AUPRC 0.60 fused; 0.64 fingersticks alone; 0.59 own rate | Missed |
| **Record as the twin's prior** | 20 held-out CGMacros participants | 0.02 mg/dL change with one day of sensor data, p = 0.16 | Missed |
| **Report from fingersticks** | 29 held-out ShanghaiT2DM patients | Ours 8.9, plain average 6.4, stale report 13.0 mg/dL | Plain baseline wins |
| **Prompt for a new wear** | 32 held-out ShanghaiT2DM recordings | AUROC 0.82 (0.63 to 0.98); plain comparison 0.82 | Descriptive, ties |
| **Readings not to believe** | Sensor lows with a fingerstick within 10 minutes | 8 of 64 confirmed; 0 of 9 at night | Descriptive |
| **The 80 % band** | 19 held-out CGMacros participants | 83.5 % held on average; 60 % to 98.5 % per patient | Descriptive |

### REPRODUCE

```
uv sync
uv run python -m chhaya.dashboard   # no dataset needed
uv run pytest -q   # takes several minutes
```

**Live demo** chhaya-digital-twin.vercel.app

**Research prototype. Not a medical device. Not for clinical decisions.**

### DOCUMENT CONTROL

| | |
|---|---|
| **Version** | 1.0 |
| **Date** | 19 October 2026 |
| **Status** | Final for submission |
| **Prepared by** | Team SynapseX, D.B.S. College, Kanpur |

### LICENCES

Code MIT. Datasets CC BY 4.0 and CC BY-NC-SA 4.0. Fonts SIL OFL 1.1.